

MULTIPLIER 2 BIT

1. Inputs and Outputs			
A 2-bit multiplier has two 2-bit inputs and a 4-bit output.			
Port	Width	MSB	LSB
A	2-bit	A[1]	A[0]
B	2-bit	B[1]	B[0]
P	4-bit	P[3]	P[0]

MAIN CODE

```
module multiplier_2bit (  
    input wire [1:0] A,  
    input wire [1:0] B,  
    output wire [3:0] P  
);  
  
    assign P = A * B;  
  
endmodule
```

TEST BENCH CODE

```
`timescale 1ns/1ps  
  
module multiplier_2bit_tb;  
  
    reg [1:0] A;  
    reg [1:0] B;  
  
    wire [3:0] P;
```

```
// Instantiate multiplier
```

```
multiplier_2bit uut (
```

```
    .A(A),
```

```
    .B(B),
```

```
    .P(P)
```

```
);
```

```
initial
```

```
begin
```

```
    //  $0 \times 0$ 
```

```
    A = 2'b00;
```

```
    B = 2'b00;
```

```
    #10;
```

```
    //  $0 \times 1$ 
```

```
    A = 2'b00;
```

```
    B = 2'b01;
```

```
    #10;
```

```
    //  $1 \times 1$ 
```

```
    A = 2'b01;
```

```
    B = 2'b01;
```

```
    #10;
```

```
    //  $1 \times 2$ 
```

```
    A = 2'b01;
```

```
    B = 2'b10;
```

```
    #10;
```

```
    //  $2 \times 2$ 
```

```
A = 2'b10;
```

```
B = 2'b10;
```

```
#10;
```

```
// 2 × 3
```

```
A = 2'b10;
```

```
B = 2'b11;
```

```
#10;
```

```
// 3 × 2
```

```
A = 2'b11;
```

```
B = 2'b10;
```

```
#10;
```

```
// 3 × 3
```

```
A = 2'b11;
```

```
B = 2'b11;
```

```
#10;
```

```
$finish;
```

```
end
```

```
initial
```

```
begin
```

```
    $monitor("Time=%0t | A=%b | B=%b | P=%b",
```

```
        $time, A, B, P);
```

```
end
```

```
endmodule
```